



# Removing Delivery Bottlenecks

Week 7 • Day 5 • DP ships today

TPM Academy



## Day 5 Objectives

- Convert top 3 queues into **3 experiments** with hypothesis / test / success criterion
- Apply the **Theory of Constraints**: one bottleneck at a time
- Run the **integration pass** across DP §§1–5
- Cross-review + sign off the DP — the 5th sibling artifact
- Declare the **Week-8 capstone candidate**



# Today's Shape

| Clock         | Block                       | Outcome                     |
|---------------|-----------------------------|-----------------------------|
| 09:00 – 09:15 | Opening                     | Pin DP §§1–4                |
| 09:15 – 10:30 | Block 1 + Activity 1        | Design 3 experiments        |
| 10:45 – 12:00 | Block 2 + Activity 2        | DP integration pass         |
| 13:00 – 14:15 | Block 3 + Activity 3        | Cross-review with rubric    |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | Sign-off + Week-8 candidate |

# Block 1 — Theory of Constraints

One bottleneck at a time



# Goldratt's Claim

A system's throughput is constrained by **exactly one bottleneck at a time**. Not many. One. Optimize anything else and you don't speed up the system — you add inventory in front of the constraint.

**Software example:** if your slowest step is "code review takes 2 days," speeding up coding doesn't help — it creates more code waiting for review.

**Find the bottleneck. Move it. Then find the next one.**



# Experiment Structure

```
### Experiment N: <name>
**Bottleneck addressed:** <queue>

<p><strong>Hypothesis:</strong> We believe that <x> will <result>
because <reasoning>.</reasoning></result></x></p>
<p><strong>Test:</strong> <what / how long / what scope></p>
<p><strong>Success criterion:</strong></p>
<ul>
<li>Baseline: <current></current></li>
<li>Target: <improved></improved></li>
<li>Window: <duration></duration></li>
</ul>
<p><strong>What could go wrong:</strong> 2-3 risks
<strong>Who runs it:</strong> named person
<strong>By when:</strong> date</p>
<p><strong>Decision after:</strong></p>
</queue></name>
```

```
<li>Met: adopt / scale / institutionalize</li>
```

- Not met: abandon / iterate / escalate

**"We'll see if it feels better"** is not an experiment.

# Activity 1 — Design 3 Experiments

Triads • 35 minutes

- 5 min: re-read top 3 queues
- 25 min: draft 3 experiments (template)
- 5 min: would your team actually try these in 60 days?



5 + 25 + 5

# Block 2 — DP Integration Pass

§§1–5 reads coherently top to bottom





# The Integration Checklist

| Check                         | Pass criterion                                    |
|-------------------------------|---------------------------------------------------|
| §1 outcomes ↔ §2 backlog      | Every output in §2 maps to an outcome in §1       |
| §2 backlog ↔ §3 tracking      | Every leading indicator corresponds to backlog    |
| §3 tracking ↔ §4 VSM          | Cycle-time leading indicators consistent with VSM |
| §4 queues ↔ §5 experiments    | Every top queue has an experiment                 |
| References to PRD/TCD/TMD/SEP | DP cites prior artifacts by section               |
| No fortune-cookie prose       | Specific to this feature                          |

## Activity 2 — Integration Pass

Triads • 40 minutes

- 15 min: solo read-through
- 10 min: pool issues
- 15 min: fix in priority order (coherence first)



15 + 10 + 15

# Block 3 — Cross-Review

The Friday DP Rubric



# The DP Rubric

| Dimension                        | Weight |
|----------------------------------|--------|
| Outcome clarity                  | 20%    |
| ADO discipline                   | 15%    |
| Output ↔ outcome pairing         | 20%    |
| Value stream realism             | 20%    |
| Bottleneck experiments           | 15%    |
| Integration with prior artifacts | 10%    |

Score thresholds:  $\geq 3.0$  ships. 2.0–2.9 with named gaps.  $< 2.0 \rightarrow$  facilitator review.

# Activity 3 — Cross-Review

Triad pairs • 40 minutes

- 5 min: pair set-up
- 20 min: cross-read with rubric
- 10 min: author triad responds
- 5 min: capture final score



5 + 20 + 10 + 5

# Block 4 — Sign-Off + Week-8 Bridge

The 5-artifact set ships; capstone candidate declared



# What Carries to Week 8

## Carries forward

- Artifact templates (PRD/TCD/TMD/SEP/DP) — in shorter form
- AI provenance discipline
- Negotiation / outcome / tracking muscles
- Cumulative AI-validation log

## New in Week 8

- **AI Spec Development** — using AI to draft technical specs that integrate the artifact set
- A real-world capstone (not FieldPulse)
- Final artifact presentations

# **Pick Your Capstone Candidate**

Each triad picks 3–5 candidates and declares a preferred one for Monday.

- **Real product / feature** from work, school, hobby, public
- **Substantive enough** for a week of work
- **Different enough** from FieldPulse to test transferability
- **Public-discussable** (no NDA conflicts)



# Activity 4 — Sign-Off + Capstone Candidate

Triads • 45 minutes • Wrap

- 10 min: adopt / defer / reject reviewer findings; sign off
- 15 min: brainstorm 3–5 capstone candidates
- 10 min: pre-flight (inputs needed; people to ask)
- 5 min: declare preferred candidate



10 + 15 + 10 + 5 • 15-min wrap



# Week 7 Wrap

## What we shipped this week

- Outcome map (DP §1)
- Loaded ADO backlog (§2)
- Tracking plan (§3)
- Value stream map (§4)
- Bottleneck experiments (§5)
- Integrated DP — 5th sibling artifact

## What opens Week 8

- **AI Spec Development & Capstone**
- Real-world capstone subject
- AI Spec dev integrated with artifact set
- Final artifact presentations Friday

**Bring to Monday:** All 5 sibling artifacts (PRD/TCD/TMD/SEP/DP) + your declared capstone candidate.

# Questions & Reflection

Which bottleneck have you been living with that you no longer accept?